

■ Math + Bit Manipulation Problems (LeetCode Style)

■ Easy Level (Basics – Start Here)

1. Print the pattern	Generate simple star/number patterns.
2. Print table of a number	Output multiplication table for a given integer.
3. Sum of first n natural numbers	Compute using formula or loop.
4. Sum of all natural numbers in range L to R	Efficient summation using formula.
5. Check if a number is even or odd	Use modulo or bitwise operator.
6. Check if a number is Positive, Negative, Zero	Simple conditional check.
7. Reverse digits of a number	Extract digits and reconstruct.
8. Sum of digits of a number	Iteratively add all digits.
9. Largest and smallest digit of a number	Find max and min among digits.
10. Print the Kth Digit	Extract specific digit position.
11. Replace all 0 with 5 in an integer	Digit manipulation problem.
12. Area of a circle	Formula: $\pi \times r^2$.
13. Check if a given year is a leap year	Leap year condition check.
14. Counting Bits	Return count of set bits for numbers 0 to n.
15. Single Number	Find unique element when others appear twice.
16. Majority Element	Find element appearing more than $n/2$ times.
17. Find Missing and Repeating Number	Detect one missing & one repeating number.
18. Add Binary	Perform binary string addition.
19. Reverse Bits	Reverse order of bits in an integer.
20. Number of 1 Bits	Count set bits (Hamming weight).
21. Palindrome Number	Check if digits form a palindrome.
22. Plus One	Increment integer represented as array of digits.
23. Sqrt(x)	Return integer square root.

■ Medium Level (Important Number Properties & Logic)

1. Series AP	Find nth term in arithmetic progression.
2. Series GP	Find nth term in geometric progression.
3. Sum of arithmetic series	Sum of first n terms in AP.
4. Sum of geometric series	Sum of first n terms in GP.
5. Binary to decimal conversion	Convert base 2 to base 10.

6. Decimal to binary conversion	Convert base 10 to base 2.
7. Decimal to octal conversion	Convert base 10 to base 8.
8. Octal to decimal conversion	Convert base 8 to base 10.
9. Binary to octal conversion	Convert base 2 to base 8.
10. Octal to binary conversion	Convert base 8 to base 2.
11. Convert a given number to words	Convert digits into English words.
12. Find all factors of a number	List all divisors of n.
13. Prime factors of a number	Factorize into primes.
14. Kth largest factor of a number	Find kth factor from largest side.
15. All divisors of a number	List every divisor systematically.
16. GCD of two numbers	Greatest common divisor (Euclidean algorithm).
17. LCM of two numbers	Least common multiple.
18. GCD of array	Find GCD of entire array.
19. Add two fractions	Perform addition with LCM denominator.
20. Find Nth root of M	Check integer root using binary search.
21. Check if number is palindrome	Number reads same forward and backward.
22. Palindromes in a given range	List palindrome numbers in interval.
23. Armstrong Numbers	Sum of digits ^{power} equals number.
24. Abundant Number	Sum of divisors > number.
25. Primitive Abundant Number	Abundant but no proper divisor is abundant.
26. Strong Number check	Sum of factorials of digits equals number.
27. Harshad (Niven) Number	Divisible by sum of its digits.
28. Minimum Flips to Make a OR b Equal to c	Bit manipulation logic problem.
29. Reverse Pairs	Count pairs with $\text{nums}[i] > 2 * \text{nums}[j]$.
30. Count Inversions	Merge sort based count of inversions.
31. Count Possible Triangles	Check triplets forming valid triangles.
32. Implement Trie (Prefix Tree)	Design insert, search, startsWith methods.
33. Search Suggestions System	Return autocomplete suggestions.
34. Non-overlapping Intervals	Erase intervals to remove overlaps.
35. Minimum Number of Arrows to Burst Balloons	Greedy interval shooting problem.
36. Single Number II	Element appearing once when others appear thrice.
37. Bitwise AND of Numbers Range	Compute AND of range [m, n].
38. Factorial Trailing Zeroes	Count zeros in factorial result.
39. Pow(x, n)	Implement power function efficiently.
40. Merge overlapping intervals	Combine intervals into one.
41. Overlapping rectangles	Check intersection between rectangles.

■ Hard Level (Advanced & Logical)

1. Factorial of a number	Compute $n!$ using loops/recursion.
2. Compute nPr	Permutations formula $n!/(n-r)!$
3. Compute nCr	Combinations formula $n!/(r!(n-r)!)$.
4. Largest prime factor	Find maximum prime factor of n .
5. Prime Number check	Efficient primality test.
6. Sieve of Eratosthenes	Efficient prime number generation.
7. Sum of all prime numbers between 1 and N	Efficient prime summation.
8. Pairs of prime numbers	Generate pairs with sum constraints.
9. Count prime numbers that can be expressed as sum of two primes	Advanced combinatorial problem.
10. Check if a number can be expressed as sum of two primes	Goldbach's conjecture type problem.
11. Jumping Numbers	Adjacent digits differ by 1.
12. Perfect Numbers	Sum of divisors equals number.
13. Pair cube count	Count pairs with cube sum equal.
14. Square Free Number	No prime factor appears more than once.
15. Fibonacci Numbers	Generate n th Fibonacci number.
16. Closest Number	Find closest divisible number.
17. Count ways to reach the n th stair	Dynamic programming problem.
18. Roots of Quadratic equation	Solve quadratic equation roots.
19. Permutations to arrange N persons around a circular table	Circular permutation count.
20. Count the triplets	Find triplets with given sum.
21. Pythagorean Triplet	Check if triplet forms right triangle.
22. Maximum possible sum of products	Arrange to maximize product sum.
23. Find duplicates in a small ranged array	Detect duplicates using marking.
24. Find Duplicate (cycle detection)	Linked list cycle detection technique.
25. Find the smallest positive integer not representable as sum of given numbers	Subset sum problem.
26. Check if given four points form a square	Distance-based validation.
27. Max Points on a Line	Find collinear maximum points.